Fiscal Year:	FY 2014	**Task Last Updated:**	FY 01/16/2014
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events (ELFSTONE)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Tufts University		
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PI Address 2:	025 Anderson Hall, 200 College Ave		
PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	09/20/2010	**End Date:**	11/30/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	6
No. of Master's Candidates:	9	**No. of Bachelor's Degrees:**	9
No. of Bachelor's Candidates:	11	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 11/30/2013 per NSSC information (previously was changed to 9/19/2014, per NSSC information at that time) (Ed., 12/26/13) NOTE: New end date is 9/19/2014, per NSSC information (Ed., 6/24/13)		
Key Personnel Changes/Previous PI:	Douglas Matson is U.S. Team Member on ESA-sponsored projects, "Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC)" and "Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS)."		
COI Name (Institution):			
Grant/Contract No.:	NNX10AV27G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Funding is for Dr. Matson's role as Team Member on the subprojects described below, associated with ESA-sponsored main projects as noted. Main Project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany Subproject: Transformation kinetics - ISS Using results from thermophysical property evaluation and combining these with the MHD convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the MSL-EML to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: Management and Coordination. Coordination of the scientific program, Coordination of the ground based measurement program, Coordination of the microgravity experiment program, Contact with the national and international space agencies. Short Duration Microgravity Measurements - PF. Conduct parabolic flights on selected alloys. Long Duration microgravity Experiments - ISS. Surface tension and viscosity as a function of temperature, Density as a function of temperature, Solidification growth velocity. Preparation of the Microgravity Experiments. Experimental protocols for processing on the ISS , Verification of flight samples for the ISS. MODELLING A: Thermodynamics, Phase Selection. Thermodynamic phase diagram calculation, Modeling of the influence of convection on phase selection.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The value of this research is that the results have application to the design of industrial welding, spray forming, and casting operations for a commercially important class of soft magnetic materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection, and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Task Progress:	Finalized parameter set development for Batch I FeCo flight alloy samples (#6 Fe45Co55 and #7 Fe50Co50) and proposed Batch II sample compositions (#32 TiAlTa and #40 Fe60Co40). Prepared flight plan for ESA Parabolic Flight September 2-13 in Merignac FR to test camera sensitivity and optimize operational settings for recalescence detection. Hosted visit of future NASA Post-doc Vijay Kumar Nov 10-12 to tour Tufts campus. Completed Post-doc training of Jonghyun Lee who moves on to a Research Faculty appointment at University of Massachusetts – Amherst. International Working Group 11-13 December 2013 at ESA-ESTEC in Noordwijk ND. IWG 9 involved evaluation of final dust limits for upcoming safety reviews, confirming sample processing order for Batch I, sample selection for Batch II, and proposing samples for Batch III. Procured new Tufts SEM to conduct in-house screening of sample microstructures. Developed protocols for EDX elemental analysis and backscatter image processing. Graduated two MS students and six BS students. Conducted planning for upcoming ESA parabolic flight February 2014. Conducted planning for flight support ESL quenching tests at ESL-MSFC March 2014. (Ed. Note 1/17/14: Continued by Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events ; grant NNX14AB74G. period of performance 12/1/2013-11/30/2015)
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Articles in Peer-reviewed Journals	Lee J, Matson DM, Binder S, Kolbe M, Herlach D, Hyers RW. "Magnetohydrodynamic modeling and experimental validation of convection inside electromagnetically levitated Co-Cu droplets." Metallurgical and Materials Transactions B. 2014 Jun;45(3):1018-23. (Reported originally as "2013 Dec. Online publication.") http://dx.doi.org/10.1007/s11663-013-9995-5 , Jun-2014